



SYDNEY GRAMMAR SCHOOL
MATHEMATICS DEPARTMENT
TRIAL EXAMINATIONS 2004

FORM VI

MATHEMATICS EXTENSION 1

Examination date

Tuesday 10th August 2004

Time allowed

2 hours (plus 5 minutes reading time)

Instructions

- All seven questions may be attempted.
- All seven questions are of equal value.
- All necessary working must be shown.
- Marks may not be awarded for careless or badly arranged work.
- Approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

Collection

- Write your candidate number clearly on each booklet.
- Hand in the seven questions in a single well-ordered pile.
- Hand in a booklet for each question, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Keep the printed examination paper and bring it to your next Mathematics lesson.

Checklist

- SGS booklets: 7 per boy. A total of 1000 booklets should be sufficient.
- Candidature: 121 boys.

Examiner

MLS

QUESTION ONE (12 marks) Use a separate writing booklet.

Marks

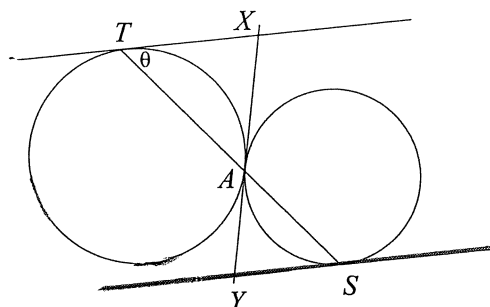
- (a) Solve the inequation $\frac{4}{5-x} \leq 1$. 3
- (b) For what value of p is the expression $4x^3 - x + p$ divisible by $x + 3$? 2
- (c) Expand $(a + \frac{1}{2})^5$, expressing each term in its simplest form. 2
- (d) Given the points $A(1, 4)$ and $B(5, 2)$, find the co-ordinates of the point that divides the interval AB externally in the ratio $1 : 3$. 2
- (e) Find $\int x(1 - x^2)^5 dx$, using the substitution $u = 1 - x^2$, or otherwise. 3

QUESTION TWO (12 marks) Use a separate writing booklet

Marks

- (a) Consider the parabola $x = 4t$, $y = 2t^2$.
- (i) Find the gradient of the parabola at the point where $t = 4$. 1
- (ii) Find the equation of the tangent to the parabola at $t = 4$. 2

(b)



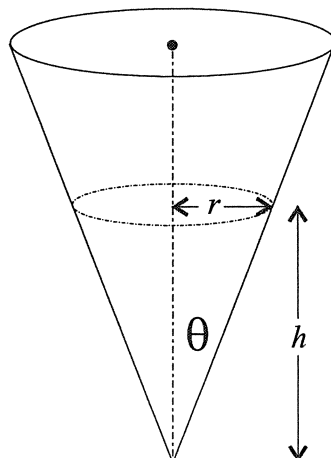
In the diagram above, two circles touch one another externally at the point A . A straight line through A meets one of the circles at T and the other at S . The tangents at T and S meet the common tangent at A at X and Y respectively.

Let $\theta = \angle XTA$.

- (i) Explain why $\angle XAT$ is θ . 1
- (ii) Prove that $TX \parallel YS$. 2
- (c) (i) Write down the first three terms in the expansion of $(1 + mx)^n$. 1
- (ii) If $(1 + mx)^n \equiv 1 - 4x + 7x^2 - \dots$, find the values of m and n . 3
- (d) Evaluate $\lim_{x \rightarrow 0} \frac{5x \cos 2x}{\sin x}$, showing your reasoning. 2

QUESTION THREE (12 marks) Use a separate writing booklet.**Marks**

(a)



The diagram above shows a container in the shape of a right circular cone. The semi-vertical angle $\theta = \tan^{-1} \frac{1}{2}$.

Water is poured in at the constant rate of 10 cm^3 per minute.

Let the height of the water at time t seconds be $h \text{ cm}$, let the radius of the water surface be $r \text{ cm}$, and let the volume of water be $V \text{ cm}^3$.

(i) Show that $r = \frac{1}{2}h$.

1

(ii) Show that $V = \frac{1}{12}\pi h^3$.

1

(iii) Find the exact rate at which h is increasing when the height of the water in the cone is 50 cm .

2

(b) Show that there is no term independent of x in the expansion of $\left(2x^2 - \frac{1}{4x}\right)^{11}$.

3

(c) Evaluate $\int_{-1}^0 x\sqrt{1+x} \, dx$, using the substitution $u = 1+x$.

4

(d) Find $\int \sin x \cos^3 x \, dx$.

1

Exam continues overleaf ...

QUESTION FOUR (12 marks) Use a separate writing booklet.

Marks

(a) If $y = \frac{1}{200}te^{-t}$, show that $\frac{dy}{dt} = \frac{1}{200}(1-t)e^{-t}$. 1

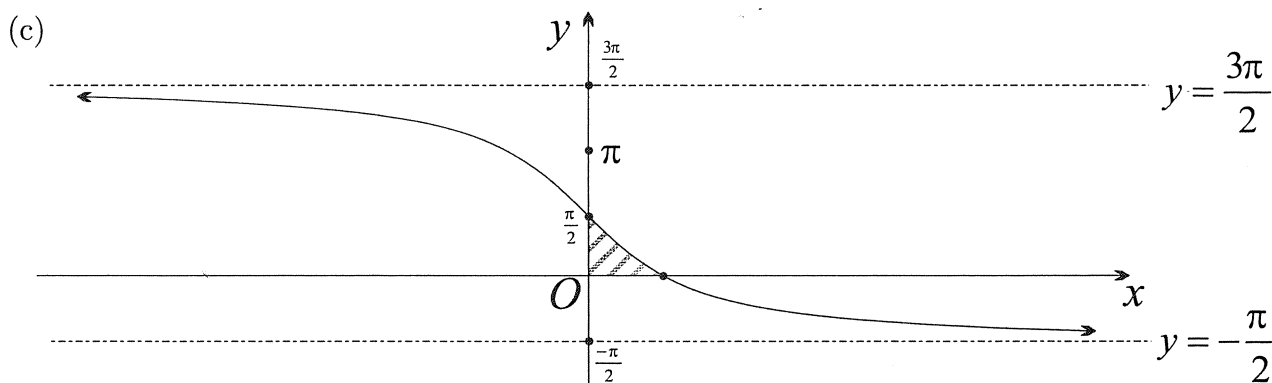
- (b) Fred has recently consumed three standard alcoholic drinks. Immediately after he has finished his last drink, his blood alcohol level is measured over a four-hour period.

Let his blood alcohol level at any time t be A , where t is the time in hours after his last drink.

It is found that the rate of change $\frac{dA}{dt}$ of his blood alcohol content is given by

$$\frac{dA}{dt} = \frac{1}{200}(1-t)e^{-t}, \text{ where } 0 \leq t \leq 4.$$

- (i) Show that his blood alcohol content increases during the first hour and decreases after the first hour. 2
- (ii) Initially his blood alcohol content was 0.0005. Find A as a function of t . You will need to use part (a). 2
- (iii) Determine his maximum alcohol content during the four-hour period. Give your answer correct to four decimal places. 1



The graph of the curve $y = \frac{\pi}{2} - 2 \tan^{-1} x$ is drawn above. It cuts the y -axis at $(0, \frac{\pi}{2})$.

- (i) Write down the domain of the inverse function of $y = \frac{\pi}{2} - 2 \tan^{-1} x$. 1
- (ii) Find the equation of the inverse function of $y = \frac{\pi}{2} - 2 \tan^{-1} x$. 1
- (iii) Find the volume generated when the shaded region is rotated about the y -axis. 4

QUESTION FIVE (12 marks) Use a separate writing booklet.

Marks

(a) Evaluate $\int_0^4 \frac{1}{3 + \sqrt{x}} dx$, using the substitution $x = (u - 3)^2$. 3

(b) (i) Write down the expansion of $(1 + x)^n$ in ascending powers of x . Then differentiate both sides of your identity. 1

(ii) Make an appropriate substitution for x to show that 1

$$\binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + 4\binom{n}{4} + \cdots + n\binom{n}{n} = n(2^{n-1}).$$

(iii) Hence find an expression for 1

$$2\binom{n}{1} + 3\binom{n}{2} + 4\binom{n}{3} + 5\binom{n}{4} + \cdots + (n+1)\binom{n}{n}.$$

(c) Find values for R and α if $\sqrt{3}\sin\theta - \cos\theta = R\cos(\theta + \alpha)$, where R and α are positive constants and $0 < \alpha < 2\pi$. 2

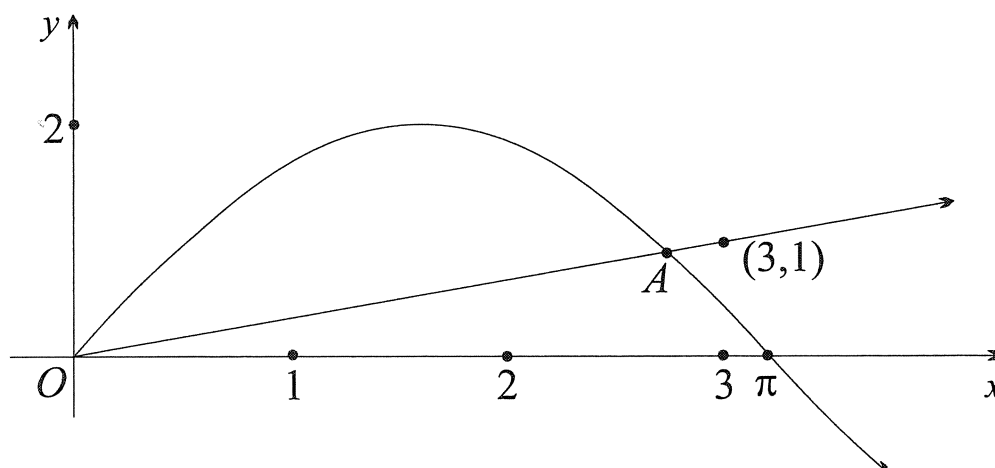
(d) Use the method of mathematical induction to prove that 4

$$\frac{1}{2!} + \frac{2}{3!} + \cdots + \frac{n}{(n+1)!} = \frac{(n+1)! - 1}{(n+1)!}, \text{ for all positive integers } n.$$

QUESTION SIX (12 marks) Use a separate writing booklet.

Marks

(a)



The sketch above shows the curve $y = 2\sin x$ and the line $x - 3y = 0$. The graphs meet at the point A in the first quadrant.

(i) Write down an equation whose solution gives the x -coordinate of A . 1

(ii) An approximate value for the x -coordinate of A is $x = 3$. Apply Newton's method once to find a closer approximation for this value. Give your answer correct to one decimal place. 2

Exam continues overleaf ...

- (b) Newton's law of cooling states that a body cools according to the equation

$$\frac{dT}{dt} = -k(T - S),$$

where T is the temperature of the body at time t , S is the temperature of the surroundings and k is a constant.

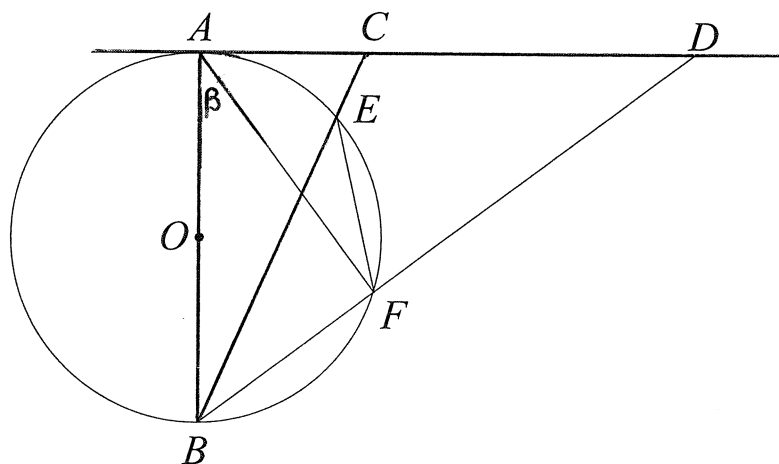
- (i) Show that $T = S + Ae^{-kt}$ satisfies the equation, where A is a constant. 1

- (ii) A metal rod has an initial temperature of 470°C and cools to 250°C in 10 minutes. The surrounding temperature is 30°C .

- (α) Find the value of A and show that $k = \frac{1}{10} \log_e 2$. 2

- (β) Find how much longer it will take the rod to cool to 70°C , giving your answer correct to the nearest minute. 2

(c)



In the diagram above, the straight line ACD is a tangent at A to the circle with centre O . The interval AOB is a diameter of the circle. The intervals BC and BD meet the circle at E and F respectively.

Let $\angle BAF = \beta$.

Copy or trace this diagram into your answer booklet.

- (i) Explain why $\angle ABF = \frac{\pi}{2} - \beta$. 1

- (ii) Prove that the quadrilateral $CDFE$ is cyclic. 3

QUESTION SEVEN (12 marks) Use a separate writing booklet.**Marks**

- (a) Car A and car B are travelling along a straight level road at constant speeds V_A and V_B respectively. Car A is behind car B , but is travelling faster.

When car A is exactly D metres behind car B , car A applies its brakes, producing a constant deceleration of $k \text{ m/s}^2$.

- (i) Using calculus, find the speed of car A after it has travelled a distance x metres under braking. 2

- (ii) Prove that the cars will collide if $V_A - V_B > \sqrt{2kD}$. 4

- (b) A particle is moving in simple harmonic motion of period T about a centre O . Its displacement at any time t is given by $x = a \sin nt$, where a is the amplitude.

- (i) Draw a neat sketch of one period of this displacement–time equation, showing all intercepts. 1

- (ii) Show that $\dot{x} = \frac{2\pi a}{T} \cos \frac{2\pi t}{T}$. 1

- (iii) The point P lies D units on the positive side of O . Let V be the velocity of the particle when it first passes through P . 4

Show that the time between the first two occasions when the particle passes through P is $\frac{T}{\pi} \tan^{-1} \frac{VT}{2\pi D}$.

END OF EXAMINATION

Question 1 2004 Trial Ex I

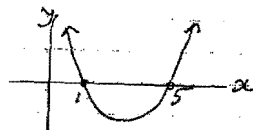
(a) $\frac{4}{5-x} \leq 1$

$4(5-x) \leq (5-x)^2$ ✓

$4(5-x) - (5-x)^2 \leq 0$

$(5-x)(4-(5-x)) \leq 0$

$(5-x)(x-1) \leq 0$ ✓



Solution: $x \leq 1$ or $x \geq 5$. ✓

(b) Let $P(x) = 4x^3 - x + p$.

If $P(x)$ is divisible by $(x+3)$ then $P(-3) = 0$

$P(-3) = -108 + 3 + p$ ✓

$-108 + 3 + p = 0$

$p = 105$ ✓

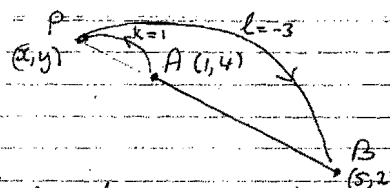
(c) $(a + \frac{1}{2})^5 = a^5 + 5a^4(\frac{1}{2}) + 10a^3(\frac{1}{2})^2 + 10a^2(\frac{1}{2})^3 + 5a(\frac{1}{2})^4 + (\frac{1}{2})^5$

$= a^5 + \frac{5}{2}a^4 + \frac{5}{2}a^3 + \frac{5}{4}a^2 + \frac{5}{16}a + \frac{1}{32}$

✓ for correct bin coeff 1, 5, 10, 10, 5, 1

✓ for all correct.

(d)



$x = \frac{lx_1 + kx_2}{l+k}$, $y = \frac{ly_1 + ky_2}{l+k}$

$= \frac{-3+5}{-2}$, $y = \frac{-12+2}{-2}$

$= -1$, $= 5$

The point is $(-1, 5)$ ✓ ✓

Give ✓ if they find the pt that divides AB internally.

(e) $\int x(1-x^2)^5 dx$

$= \int -\frac{1}{2} u^5 du$ ✓

$= -\frac{1}{2} \frac{u^6}{6} + c$

$= -\frac{1}{12} (1-x^2)^6 + c$ ✓

$u = 1-x^2$

$du = -2x dx$ ✓

$-\frac{1}{2} du = x dx$

Question 2

(i) (1) $x = 4t$ and $y = 2t^2$
 $\frac{dx}{dt} = 4$ $\frac{dy}{dt} = 4t$

so $\frac{dy}{dx} = \frac{4t}{4}$

$= t$

so at $t=4$, the gradient is 4 ✓

(ii) When $t=4$, $x=16$, $y=32$ ✓

Tangent equation is $y-32 = 4(x-16)$

$y-32 = 4x-64$

$y = 4x-32$ ✓

(b)

(i) Tangents to a circle from an external point are equal ✓

$\therefore TX = XA$

So $\triangle TXA$ is isosceles and $\angle XTA = \angle XAT = \theta$
 (base angles of isosceles triangle)

(ii) Similarly, $\triangle AYS$ is isosceles with base angles $\angle YAS$ and $\angle YSA$ equal }
 But $\angle YAS = \angle TTX = \theta$ (vertically opposite)
 So $\angle YSA = \theta$ and $\angle XTA = \theta$
 But these are alternate }
 So $TX \parallel YS$ ✓

(c) (i) $(1+mx)^n = 1 + nmx + \frac{n(n-1)}{2}(mx)^2 + \dots$ ✓

(ii) $1-4x+7x^2-\dots = 1 + nmx + \frac{n(n-1)}{2}(mx)^2 + \dots$

equate coefficients of x : $-4 = nm$ ①

equate coefficients of x^2 : $7 = \frac{n(n-1)}{2}m^2$ ② ✓

From ①, $m = -\frac{4}{n}$, substitute this in ②.

$14 = n(n-1) \frac{16}{n^2}$

$14 = \frac{16(n-1)}{n}$

$14n = 16n-16$

$2n = 16$

$n = 8$

$m = -\frac{4}{8}$

$= -\frac{1}{2}$

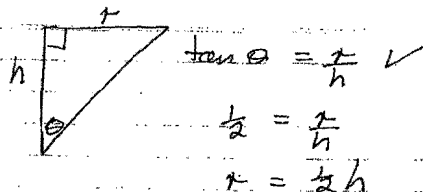
$n=8$ and $m=-\frac{1}{2}$ ✓

✓ for a sensible method

(d) $\lim_{x \rightarrow 0} \frac{5x \cos 2x}{\sin 2x} = \lim_{x \rightarrow 0} \frac{5x}{\sin 2x} \times \lim_{x \rightarrow 0} \cos 2x$
 $= 5 \lim_{x \rightarrow 0} \frac{x}{\sin x} \times 1$ ✓
 $= 5 \times 1$
 $= 5$ ✓

Q3

a) (i)



$$\tan \theta = \frac{r}{h} \quad \checkmark$$

$$\frac{1}{2} = \frac{r}{h}$$

$$r = \frac{1}{2}h$$

$$(ii) \quad V = \frac{1}{3}\pi r^2 h$$

$$= \frac{1}{3}\pi \left(\frac{1}{2}h\right)^2 h \quad \checkmark$$

$$= \frac{1}{12}\pi h^3$$

$$(iii) \quad \text{Find } \frac{dh}{dt}, \text{ given } \frac{dV}{dt} = 10$$

$$\frac{dh}{dt} = \frac{dh}{dV} \frac{dV}{dt} \quad \checkmark$$

$$\text{now } V = \frac{1}{12}\pi h^3$$

$$\frac{dV}{dh} = \frac{\pi h^2}{4} \text{ so } \frac{dh}{dV} = \frac{4}{\pi h^2}$$

$$\text{so } \frac{dh}{dt} = \frac{4}{\pi h^2} \times 10$$

$$= \frac{4}{\pi \times 50^2} \times 10 \text{ when } h = 50$$

$$= \frac{40}{\pi \times 50 \times 50} \quad \checkmark$$

$$= \frac{4}{250\pi} \text{ cm per minute}$$

(b)

$$\text{The general term is } \binom{22}{r} (2x^2)^{22-r} (-4x)^r \quad \checkmark$$

The index of x is $22 - 2r + r = 0$ if the term is independent of x .

$$22 - r = 0$$

$$r = 22$$

Since r must be an integer, there is no term independent of x . $\checkmark$

(c)

$$\int_{-1}^0 x\sqrt{1+x} \, dx$$

$$u = 1+x$$

$$du = dx$$

$$\text{when } x=0, u=1$$

$$\text{when } x=-1, u=0$$

$$= \int_0^1 (u-1)u^{\frac{1}{2}} du \quad \checkmark$$

$$= \int_0^1 (u^{\frac{3}{2}} - u^{\frac{1}{2}}) du \quad \checkmark$$

$$= \left[\frac{2u^{\frac{5}{2}}}{5} - \frac{2u^{\frac{3}{2}}}{3} \right]_0^1 \quad \checkmark$$

$$= \left(\frac{2}{5} - \frac{2}{3} \right) - (0)$$

$$= -\frac{4}{15} \quad \checkmark$$

cd)

$$\int \sin x \cos^3 x \, dx$$

$$= -\frac{1}{4} \cos^4 x + c \quad \checkmark$$

Question 4

a) $y = \frac{1}{200} t e^{-t}$

$$\frac{dy}{dt} = \frac{1}{200} [t \times (-e^{-t}) + e^{-t}] \quad \checkmark$$

$$= \frac{1}{200} e^{-t} (1-t)$$

b) (i) $\frac{dA}{dt} = \frac{1}{200} (1-t) e^{-t}$

Now, $\frac{1}{200} (1-t) e^{-t} > 0$ when $1-t > 0$ or $t < 1$ ✓

so, for $0 < t < 1$, $\frac{dA}{dt} > 0$ and A is

increasing.

and $\frac{1}{200} (1-t) e^{-t} < 0$ when $1-t < 0$ or $t > 1$ ✓

so, for $t > 1$, $\frac{dA}{dt} < 0$ and A is

decreasing.

(ii) $t=0$, $A=0.0005$

$$A = \int \frac{1}{200} (1-t) e^{-t} dt$$

$$= \frac{1}{200} t e^{-t} + c \quad \text{from (a).} \quad \checkmark$$

when $t=0$, $0.0005 = 0 + c$ so $c = 0.0005$

$$A = \frac{1}{200} t e^{-t} + 0.0005 \quad \checkmark$$

(iii) From (i), the maximum A is when $t=1$.

so, if $t=1$, $A = \frac{1}{200} e^{-1} + 0.0005$

$$= 0.001839 + 0.0005 \quad \checkmark$$

$$= 0.002339$$

$$= 0.0023$$

(c) (i) $-\frac{\pi}{2} < x < \frac{3\pi}{2}$ ✓

(ii) $x = \frac{\pi}{2} - 2 \tan^{-1} y$

$$2 \tan^{-1} y = \frac{\pi}{2} - x$$

$$\tan^{-1} y = \frac{1}{2} \left(\frac{\pi}{2} - x \right) \quad \checkmark$$

$$y = \tan \left(\frac{\pi}{4} - \frac{1}{2} x \right), \quad -\frac{\pi}{2} < x < \frac{3\pi}{2}$$

(iii) $V = \pi \int x \, dy$

or, using the inverse function

$$V = \pi \int y^2 \, dx$$

$$= \pi \int_0^{\frac{\pi}{2}} \tan^2 \left(\frac{\pi}{4} - \frac{1}{2} x \right) dx \quad \checkmark$$

$$= \pi \int_0^{\frac{\pi}{2}} \sec^2 \left(\frac{\pi}{4} - \frac{1}{2} x \right) - 1 \, dx \quad \checkmark$$

$$= \pi \left[-2 \tan \left(\frac{\pi}{4} - \frac{1}{2} x \right) - x \right]_0^{\frac{\pi}{2}} \quad \checkmark$$

$$= \pi \left[\left(-2 \tan 0 - \frac{\pi}{2} \right) - \left(-2 \tan \frac{\pi}{4} - 0 \right) \right] \quad \checkmark$$

question 5

$$1 \int_0^4 \frac{1}{3+\sqrt{x}} dx$$

$$x = (u-3)^2$$

$$dx = 2(u-3) du$$

when $x=4$, $u=5$
when $x=0$, $u=3$.

$$= \int_3^5 \frac{2(u-3)}{u} du$$

$$= 2 \int_3^5 (1 - \frac{3}{u}) du$$

$$= 2 [u - 3 \log_e u]_3^5$$

$$= 2 [(5 - 3 \log 5) - (3 - 3 \log 3)]$$

$$= 2 (2 - 3 \log 5 + 3 \log 3)$$

$$= 2 (2 + \log \frac{3}{5})$$

$$(b) (1+x)^n = 1 + \binom{n}{1}x + \binom{n}{2}x^2 + \binom{n}{3}x^3 + \dots + \binom{n}{n}x^n$$

$$n(1+x)^{n-1} = \binom{n}{1} + 2\binom{n}{2}x + 3\binom{n}{3}x^2 + \dots + n\binom{n}{n}x^{n-1}$$

iii let $x=1$

$$LHS = n(2)^{n-1}$$

$$RHS = \binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + \dots + n\binom{n}{n}$$

iv let $x=1$ in expansion of $(1+x)^n$

$$2^n = 1 + \binom{n}{1} + \binom{n}{2} + \binom{n}{3} + \dots + \binom{n}{n}$$

$$\text{and } n2^{n-1} = \binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + \dots + n\binom{n}{n}$$

$$\text{adding } 2^n + n2^{n-1} = 1 + 2\binom{n}{1} + 3\binom{n}{2} + 4\binom{n}{3} + \dots + (n+1)\binom{n}{n}$$

$$\text{so } 2^n + n2^{n-1} - 1 = 2\binom{n}{1} + 3\binom{n}{2} + 4\binom{n}{3} + \dots + (n+1)\binom{n}{n}$$

$$(c) \sqrt{3} \sin \theta - \cos \theta = R \cos(\theta + \alpha)$$

$$= R \cos \theta \cos \alpha - R \sin \theta \sin \alpha$$

$$\text{so } -1 = R \cos \alpha \quad \text{and} \quad \sqrt{3} = -R \sin \alpha$$

$$R = \sqrt{3+1}$$

$$= 2$$

$$\text{and } \alpha = \frac{4\pi}{3}$$

(d)

If $n=1$

$$LHS = \frac{1}{2!} = \frac{1}{2}$$

$$RHS = \frac{2!-1}{2!}$$

$$= \frac{1}{2}$$

so the statement is true when $n=1$.

Now suppose the statement is true for some value of n , k .

$$\text{i.e. } \frac{1}{2!} + \frac{2}{3!} + \dots + \frac{k}{(k+1)!} = \frac{(k+1)!-1}{(k+1)!}$$

We now prove the result for $n=k+1$

$$\text{That is prove that } \frac{1}{2!} + \frac{2}{3!} + \dots + \frac{k}{(k+1)!} + \frac{k+1}{(k+2)!} = \frac{(k+2)!-1}{(k+2)!}$$

$$\text{Now, LHS} = \frac{(k+1)!-1}{(k+1)!} + \frac{k+1}{(k+2)!} \quad \text{using induction hypothesis}$$

$$LHS = \frac{(k+1)! - 1}{(k+1)!} + \frac{k+1}{(k+2)!}$$

$$= \frac{(k+2)(k+1)! - (k+2) + (k+1)}{(k+2)!}$$

$$= \frac{(k+2)! - 1}{(k+2)!}$$

$$= RHS.$$

So, the statement is true for $k+1$ as long as it is true for k .
Hence, by the principle of mathematical induction, it is true for all positive integers n .

Question 6.

(a) i) At A , $y = 2 \sin x$ and $y = \frac{1}{3}x$

so we want $2 \sin x = \frac{1}{3}x$

or $2 \sin x - \frac{1}{3}x = 0$

ii) Let $f(x) = 2 \sin x - \frac{1}{3}x$
 $f'(x) = 2 \cos x - \frac{1}{3}$

$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$, $x_0 = 3$

$= 3 - \frac{2 \sin 3 - 1}{2 \cos 3 - \frac{1}{3}}$

$= 3 - 0.3102$

$= 2.7$

(b) (i) $T = S + Ae^{-kt}$
 $\frac{dT}{dt} = -kAe^{-kt}$, $Ae^{-kt} = T - S$
 $= -k(T - S)$

(i) (a) $T = 30^\circ + Ae^{-kt}$
when $t=0$, $47.0^\circ = 30^\circ + Ae^0$
 $A = 17.0^\circ$

when $t=10$, $25.0 = 30 + 17.0e^{-10k}$
 $17.0e^{-10k} = -2.0$
 $e^{-10k} = -\frac{1}{8.5}$

$-10k = \log_e -\frac{1}{8.5}$
 $k = -\frac{1}{10} \log_e -\frac{1}{8.5}$
 $= \frac{1}{10} \log_e 8.5$

(3) Find t when $T = 70^\circ$
 $70 = 30 + 40e^{-kt}$ $k = \frac{1}{10} \ln 2$
 $e^{-kt} = \frac{40}{110}$
 $-kt = \log_e \frac{4}{11}$
 $t = \frac{\log_e \frac{11}{4}}{-\frac{1}{10} \log_e 2}$
 $\approx 35 \text{ min}$

(c) (i) $\angle AFB = \frac{\pi}{2}$ (the angle in a semicircle is a right angle).
 So $\angle ABF = \pi - (\frac{\pi}{2} + \beta)$ (the angle sum of $\triangle AFB$ is π)
 $= \frac{\pi}{2} - \beta$

(ii) $\angle BAD = \frac{\pi}{2}$ (angle between tangent and radius is $\frac{\pi}{2}$).
 So $\angle ADB = \pi - (\frac{\pi}{2} + (\frac{\pi}{2} - \beta))$ (angle sum of $\triangle ADB$ is π).
 $= \beta$

Now, $\angle BAF = \angle BEF$ (both subtended at the circumference by arc BF)
 $= \beta$

So, $\angle BEF = \angle CDF$
 So, CDEF is cyclic (exterior angle equals interior opposite angle)

Question 2.

(a) (i) For car A
 $\ddot{x} = -k$, since the car is decelerating
 $\frac{1}{2}v^2 = -kx + c$

When $x=0$, $v=V_A$
 so $\frac{1}{2}V_A^2 = 0 + c$ making $c = \frac{1}{2}V_A^2$
 $v^2 = -2kx + V_A^2$
 and speed $= \sqrt{V_A^2 - 2kx}$

(ii) For car A:

$\ddot{x} = -k$
 Integrating, $\dot{x} = -kt + c_1$
 When $t=0$, $\dot{x} = V_A$ making $c_1 = V_A$
 So $\dot{x} = -kt + V_A$
 Integrating, $x = -\frac{1}{2}kt^2 + tV_A + c_2$
 When $t=0$, $x=0$, taking the origin of displacement at car A, so $c_2 = 0$
 We have $x = tV_A - \frac{1}{2}kt^2$

For car B:

$\ddot{x} = 0$
 Integrating, $\dot{x} = c_3$
 When $t=0$, $\dot{x} = V_B$ making $c_3 = V_B$
 So $\dot{x} = V_B$
 Integrating, $x = tV_B + c_4$
 When $t=0$, $x=D$, car B is D metres in front of car A, making $c_4 = V_B$
 So $x = tV_B + D$

When the cars collide, their displacements are equal, so we have
 $tV_B + D = tV_A - \frac{1}{2}kt^2$

this is a quadratic in t .
For t to have a real value, the discriminant must be positive

$$\frac{1}{2}kt^2 - tV_A + tV_B + D = 0$$

$$kt^2 - 2t(V_A - V_B) + 2D = 0$$

$$\Delta = 4(V_A - V_B)^2 - 8kD$$

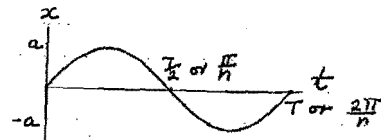
$$4(V_A - V_B)^2 - 8kD > 0$$

$$(V_A - V_B)^2 > 2kD$$

$$V_A - V_B > \sqrt{2kD}$$

since $V_A > V_B$ and so $V_A - V_B$ is positive:

(b) (i)



$$T = \frac{2\pi}{n} \text{ so } n = \frac{2\pi}{T}$$

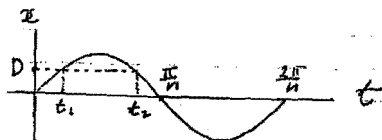
(ii)

$$x = a \sin nt$$

$$\dot{x} = na \cos nt$$

$$= \frac{2\pi a}{T} \cos \frac{2\pi}{T} t$$

(iii)



At P we have $D = a \sin \frac{2\pi}{T} t$ ①

And $V = \frac{2\pi a}{T} \cos \frac{2\pi}{T} t$ ②

$$\textcircled{1} \div \textcircled{2} \quad \frac{D}{V} = \frac{aT}{2\pi a} \tan \frac{2\pi t}{T} \quad \checkmark$$

$$\frac{D}{VT} = \tan \frac{2\pi t}{T}$$

Let t_1 and t_2 be the first two times upon the particle is at P.
Then $\frac{2\pi t_1}{T} = \tan^{-1} \frac{2\pi D}{VT}$

$$t_1 = \frac{T}{2\pi} \tan^{-1} \frac{2\pi D}{VT}$$

And $t_2 = \frac{T}{2} - \frac{T}{2\pi} \tan^{-1} \frac{2\pi D}{VT} \quad \checkmark$

So the difference in times is

$$t_2 - t_1 = \frac{T}{2} - \frac{2T}{2\pi} \tan^{-1} \frac{2\pi D}{VT}$$

$$= \frac{T}{\pi} \left(\frac{\pi}{2} - \tan^{-1} \frac{2\pi D}{VT} \right) \quad \checkmark$$

$$= \frac{T}{\pi} \tan^{-1} \frac{VT}{2\pi D}, \text{ using}$$

complementary angles.

